RadixMaths

A Level Mathematics

Practice Paper 2

Pure Mathematics

Level: **Hard**

Similar in difficulty to the hardest real A Level pure papers.

Time: 2 hours **Total marks: 100**

For students of every exam board. This paper covers only the pure mathematics that the Department for Education requires in every A Level Mathematics course, so it suits AQA (7357), Edexcel (9MA0), OCR A (H240) and OCR B MEI (H640). Questions may use content from Year 12 or Year 13.

You will need: a calculator, your exam board's formulae booklet, and paper to write on.

Instructions

- Answer **all** the questions, and label your answers to each part clearly.
- Show enough working to make your method clear. Answers without working may not gain full credit.
- Where a question says *show detailed reasoning*, or says that solutions relying on calculator technology are not acceptable, you must show every step of your working.
- Give inexact answers to three significant figures unless the question says otherwise. Use radians unless a question uses degrees.
- Diagrams are not drawn to scale.

Information

- The total mark for this paper is 100. The marks for each part are shown in brackets, for example **(3)**. Use them as a guide to how long to spend.
- The mark scheme and full worked solutions are in a separate document. Try the whole paper under timed conditions before you look at them.

1. *In this question you must show detailed reasoning. Solutions relying entirely on calculator technology are not acceptable.*

Solve the equation

$$2^{2x+1} - 9\,(2^x) + 4 = 0$$

(3)

(Total for Question 1 is 3 marks)

2. (a) Find the binomial expansion of

$$(8 - 3x)^{\frac{1}{3}}$$

in ascending powers of x, up to and including the term in x^2. Give each coefficient as a simplified fraction. **(4)**

(b) State the range of values of x for which the expansion is valid. **(1)**

(c) Use your expansion with $x = 0.01$ to find an approximation for $\sqrt[3]{7.97}$, giving your answer to 6 decimal places. **(1)**

(Total for Question 2 is 6 marks)

3. Prove, from first principles, that the derivative of $\cos x$ is $-\sin x$.

You may assume the formula for $\cos(A \pm B)$ and that, as $h \to 0$,

$$\frac{\sin h}{h} \to 1 \qquad \text{and} \qquad \frac{\cos h - 1}{h} \to 0$$

(5)

(Total for Question 3 is 5 marks)

4. A geometric series has first term a and common ratio r, where r is real. The sum to infinity of the series is 12. The sum of the first three terms of the series is $\frac{21}{2}$.

(a) Find the value of r and the value of a. **(3)**

(b) Find the smallest value of n for which the sum of the first n terms differs from the sum to infinity by less than 0.001. **(2)**

(Total for Question 4 is 5 marks)

5. The function f is defined by

$$\mathrm{f}(x) = \frac{2x - 1}{x^2 + 2}, \qquad x \in \mathbb{R}$$

(a) Show that

$$\mathrm{f}'(x) = \frac{-2(x - 2)(x + 1)}{(x^2 + 2)^2}$$

(3)

(b) Find the coordinates of the stationary points of the curve $y = \mathrm{f}(x)$. **(3)**

(c) Hence find the range of f. **(2)**

(Total for Question 5 is 8 marks)

6. In the triangle OAB, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

- The point P lies on OB such that $OP : PB = 2 : 1$.
- The point M is the midpoint of AB.
- The lines AP and OM intersect at the point Q.

Use a vector method to find the ratio $OQ : QM$. **(5)**

(Total for Question 6 is 5 marks)

7.

$$\mathrm{f}(x) = 3|x - 2| - 4, \qquad x \in \mathbb{R}$$

(a) Sketch the graph of $y = \mathrm{f}(x)$, showing the coordinates of the vertex and of any points where the graph meets the coordinate axes. **(3)**

(b) Solve the equation

$$\mathrm{f}(x) = \tfrac{1}{2}x + 1$$

(3)

(c) Find the set of values of the constant k for which the equation

$$\mathrm{f}(x) = kx$$

has exactly two real solutions. **(4)**

(Total for Question 7 is 10 marks)

8. (a) Express $5\sin\theta - 12\cos\theta$ in the form $R\sin(\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. Give the exact value of R and the value of α to 3 decimal places. **(3)**

(b) Solve, for $0 \leqslant \theta < 2\pi$, the equation

$$5\sin\theta - 12\cos\theta = 6.5$$

giving your answers to 2 decimal places. **(3)**

(c)

$$\mathrm{g}(\theta) = \frac{1}{20 + 5\sin\theta - 12\cos\theta}$$

Find the maximum value of $\mathrm{g}(\theta)$, and the smallest positive value of θ at which it occurs. **(3)**

(Total for Question 8 is 9 marks)

9. Use the substitution $u = \mathrm{e}^x + 1$ to show that

$$\int_0^{\ln 3} \frac{\mathrm{e}^{2x}}{\mathrm{e}^x + 1}\,\mathrm{d}x = 2 - \ln 2$$

(5)

(Total for Question 9 is 5 marks)

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10. The curve C has equation

$$x^3 + y^3 = 6xy$$

(a) Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of x and y. **(3)**

(b) Find the exact coordinates of the point on C, other than the origin, at which the tangent to C is parallel to the x-axis. **(3)**

(Total for Question 10 is 6 marks)

11. (a) Prove that

$$\frac{1 - \cos 2\theta}{\sin 2\theta} \equiv \tan\theta, \qquad \theta \neq \frac{n\pi}{2},\ n \in \mathbb{Z}$$

(3)

(b) *In this question you must show detailed reasoning. Solutions relying entirely on calculator technology are not acceptable.*

Hence solve, for $0 < x < \pi$, the equation

$$\frac{1 - \cos 4x}{\sin 4x} = 3\tan x$$

(4)

(Total for Question 11 is 7 marks)

12. (a) Use the trapezium rule with 4 strips of equal width to find an estimate for

$$\int_1^3 \ln x \, \mathrm{d}x$$

giving your answer to 4 significant figures. **(3)**

(b) Use integration by parts to find the exact value of $\displaystyle\int_1^3 \ln x \, \mathrm{d}x$. **(3)**

(c) Explain why the trapezium rule gives an underestimate for this integral. **(1)**

(Total for Question 12 is 7 marks)

13. Prove by contradiction that there are no positive integers x and y such that

$$x^2 - y^2 = 10$$

(5)

(Total for Question 13 is 5 marks)

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14. The number of fish, P, in a lake t years after the fish were first introduced is modelled by the differential equation

$$\frac{dP}{dt} = \frac{P(500 - P)}{2500}$$

Initially there were 50 fish in the lake.

(a) Solve the differential equation to show that

$$P = \frac{500}{1 + 9e^{-0.2t}}$$

(6)

(b) Find the time it takes for the number of fish to reach 250, giving your answer in years to 2 decimal places. **(2)**

(c) State what happens to the number of fish in the long term, according to the model. **(1)**

(Total for Question 14 is 9 marks)

15. Figure 2 shows the curve C with parametric equations

$$x = t^2, \qquad y = t^3 - 3t, \qquad t \in \mathbb{R}$$

The curve crosses itself at the point A on the x-axis, forming a loop. The finite region R, shown shaded, is enclosed by the loop.

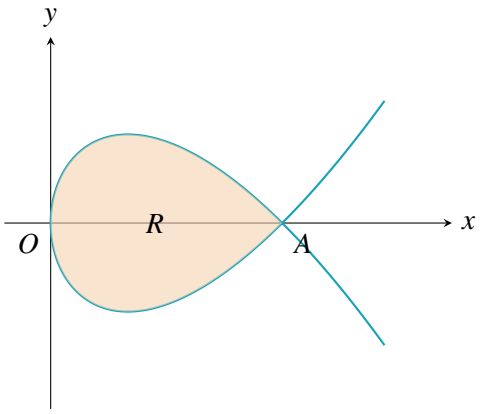


Figure 2

(a) Find the coordinates of the points on C at which the tangent is parallel to the x-axis. **(3)**

(b) Show that a Cartesian equation for C is

$$y^2 = x(x - 3)^2$$

(2)

(c) Hence find the exact area of R. **(5)**

(Total for Question 15 is 10 marks)

TOTAL FOR PAPER IS 100 MARKS

END OF QUESTIONS